

List of Projects:

qi26_1

Benchmarking QSVT-Based Matrix Inversion with Structure-Aware Block Encodings

Mentor(s): Viraj Dsouza

qi26_2

Benchmarking Efficient LCU Decompositions of Laplacian matrices for VQLS

Mentor(s): Viraj Dsouza

qi26_3

Resource-Aware Quantum Routing in Quantum Networks

Mentor(s): Haris Ansari

qi26_4

Design of Quantum-Anamorphic Encryption Against Coercive Surveillance

Mentor(s): Soumyadev Maity; Deep Raman

qi26_5

Simulated Entanglement-Inspired Communication for Federated Quantum Machine Learning Systems

Mentor(s): Ioannis Theodonis; Nouhaila Innan

qi26_6

Solving routing problems using quantum annealing

Mentor(s): Paweł Góra

qi26_7

QTagger+: Quantum Machine Learning for Ransomware Tagging and Classification with Generative Data Augmentation

Mentor(s): Simranjit Singh, Mohit Sajwan

qi26_8

Heterogeneity-Aware QUBO Formulation for Task Mapping in Network-on-Chip Architectures via Quantum Annealing

Mentor(s): Selma Dilek

qi26_9

Ground-State Energy Estimation of spin-1/2 antiferromagnetic Heisenberg Model by Hamiltonian Calibration through Hardware-Efficient VQE

Mentor(s): Muhammad Ahsan

qi26_10

Quantum Framework for Solving Nonlinear Partial Differential Equations in Physical Systems

Mentor(s): Yasemin Poyraz Koçak, Dr. Hüseyin Yıldız

qi26_11

Secure SDN Traffic Management using Post-Quantum Cryptography

Mentor(s): Abhishek Narwaria

qi26_12

QuantumSentinel: A Rigorous Benchmark of Variational Quantum Neural Networks for Zero-Day and Adversarial Intrusion Detection in IoT Networks

Mentor(s): Mohit Sajwan

qi26_13

Quantum-Native Optimization for Risk-Sensitive URLLC in UAV-RIS Networks

Mentor(s): Vieeralingaam G

qi26_14

Simulation tool for orchestrating quantum programs in distributed quantum environments.

Mentor(s): Abhishek Sawaika

qi26_15

Quantum Correlations and Bell Nonlocality in Continuous-Variable Gaussian Systems

Mentor(s): Khalid TALBI

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Accelerating hybrid quantum-classical machine learning tasks on HPC platforms.

Mentor(s): Maha A. Metawei

qi26_17

Quantum Reservoir Memory for Continuous-Time Temporal Graph Learning

Mentor(s): Krishna Bhatia; Shalini Devendrababu

qi26_18

Mixed-State g-Purity: Purification Consistency, Monotonicity, and Trade-offs

Mentor(s): Oxana Shaya

qi26_19

Noise-Adaptive Decomposition of Multi-Qubit Gates for IBM Quantum Devices

Mentor(s): Aarav Ratna

qi26_20

Quantum Computing for Combinatorial Assignment Optimization

Mentor(s): Iván Barrientos Salas

qi26_21

Quantum App for Nuclear Physics Integrated into the QCOL Open Course Ecosystem

Mentor(s): Franci Laffita Camargo

qi26_22

Dynamic Entanglement Weighted Pruning for QFL-Based Supply Chain Risk Unlearning

Sumit Chongder

qi26_23

QIOpt4QNet: Hybrid Quantum-Inspired Optimization for Fidelity-Constrained Entanglement Routing and Memory Scheduling in Quantum Networks

Mentor(s): Krishna Bhatia; Shalini Devendrababu

qi26_24

Quantum Reinforcement Learning for Decision Making

Mentor(s): Adithya Phani Thota

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Hybrid QML and Quantum String Algorithms for Efficient Document Extraction and Processing

Mentor(s): Potluri Krishna Priyatham

qi26_26

Benchmarking Quantum Algorithms in the Cloud: A Comparative Study of Classical vs. Simulated Quantum Approaches on AWS

Mentor(s): Joselyn Lagunas